



# HARDWARE CAPSTONE: DESIGN A USEFUL GADGET



### **PRESSURE COOKER**

Cooks food faster using  
steam pressure – saves time  
and fuel!



# WHAT IS A GADGET?

### **PENCIL BOX**

Holds fabric together  
safely – no more  
falling badges!



### **SAFETY PIN**

Keeps pencils, erasers and  
sharpeners organised in  
compartments!



# GADGETS SOLVE PROBLEMS



## IDLI STAND

Stacks steaming plates to  
cook many idlis at once  
without taking  
extra space!



## CYCLE PUMP

Keeps homemade food  
warm while delivering it  
across the city!

## Dabba Delivery System

Fills bicycle tyres easily so  
you can ride smoothly  
anywhere!



# THE DESIGN LOOP

Design means THINK about problems, MAKE gadgets, and SHARE your idea. This loop helps you create amazing gadgets that solve real problems!

THINK → MAKE → SHARE

Keep looping to improve your design!



Source: AI-generated data. Replace or verify before use.

# THE LEAKY WATER BOTTLE PROBLEM

## RAVI'S PROBLEM

Ravi's water bottle leaks and wastes water every day at school. The water drips into his bag and makes his books wet!

## HOW CAN HE FIX IT?

Can you think of ways Ravi could fix his leaky bottle? What materials might help?





# MANSUKHBHAI PRAJAPATI'S MITTICOOL CLAY FRIDGE

AN AMAZING INDIAN

INNOVATOR

Mansukhbhai made a fridge from clay to keep food cool without electricity! He noticed that many families in villages had no power, so he created the Mitticool fridge using traditional pottery skills. Now families can keep their vegetables and milk fresh!



# YOUR TURN: MY PROBLEM STATEMENT

## FILL YOUR WORKSHEET

What problem do you want to solve? Think about something that bothers you or someone you know.

## WORKSHEET FIELDS TO COMPLETE

- I noticed a problem when...
- The problem is...
- This bothers...
- I think a gadget could help because...





# PROBLEMS YOU CAN SOLVE



## READING TROUBLE

Grandmother can't read  
small print on medicine  
bottles.



## Wasting Water

School water tap drips  
and wastes water  
every day.

## HEAVY BAGS

Heavy school  
bags hurt when  
climbing stairs.





# NEXT STEP: MAKE YOUR GADGET

Get Ready to Build!  
Soon, you will build your gadget  
to solve your problem. Use  
simple materials like cardboard,  
tape, string, and recycled  
items to bring your idea to life!



# 3-2-1 EXIT



## 3) GADGETS

Name 3 gadgets you  
learned about today!



## 2) DESIGN LOOP STEPS

Share 1 word about how  
you feel after today's  
session!

## 1) FEELING WORD

Which 2 steps of THINK  
MAKE → SHARE do you  
remember?



# TAKE-HOME CHALLENGE

- Look around your home or school. Find one problem you want to solve!
- Notice things that don't work well or could be better. Maybe a door that squeaks, a shelf that's too high, or something that spills easily.
- You are a problem-finder now! Designers see problems everywhere.







THANK YOU FOR  
JOINING!  
YOU ARE ALREADY A  
DESIGNER!